Question 1 — 2 pts

The broadcast of radio is an example of which serial communication mode.

- ○ Half-Duplex
- ○ Full-Duplex
- ◉ Simplex
- ○ None of the above

Question 2 — 2 pts

The upload and download data streams for the Internet between the Internet Service Provider and your home/apartment/dorm Internet connection makes it an example of which serial communication mode.

- ◉ Full Duplex
- ○ Half Duplex
- ○ Simplex
- ○ None of the above

Question 3

2 pts

The COM port for IBM and Compatibles uses asynchronous serial communication.

- (●) True
- ( ) False

Question 4

5 pts

Suppose that the interrupt sources Timer 0 overflow, external interrupt 1, external interrupt 2, USART Receive Complete interrupt, and Timer 2 Clear Timer on Compare Match are enabled (ATmega32).

What is the order that these interrupt sources are serviced if they occur at the same time? Match the interrupt source to the order (1 is first, 5 is last) that the interrupt source is serviced.

1	External Interrupt 1 (INT1)
2	External Interrupt 2 (INT2)
3	Timer 2 Clear Timer on Comp
4	Timer 0 Overflow
5	USART Receive Complete

Question 5

10 pts

What are the **interrupt flags** that are used/monitored for each of the five interrupt sources below to determine if it (interrupt source) needs to be serviced? Match the interrupt sources to their associated interrupt flags.

Interrupt sources (ATmega32): Timer 0 overflow, external interrupt 1 (falling edge), external interrupt 2 (rising edge), USART Receive Complete interrupt, and Timer 2 Clear Timer on Compare Match

Timer 0 Overflow	TOV0
Timer 2 Clear Timer on Compare Match	OCF2
External Interrupt 1 (falling edge)	INTF1
USART Receive Complete	RXC
External Interrupt 2 (rising edge)	INTF2
USART UDR Empty	UDRE

Question 6

8 pts

What I/O register(s) are the interrupt flags contained for the following interrupt sources?

Interrupt sources (ATmega32): Timer 0 overflow, external interrupt 1 (falling edge), external interrupt 2 (rising edge), USART Receive Complete interrupt, and Timer 2 Clear Timer on Compare Match.

USART UDR Empty	UCSRA
Timer 0 Overflow	TIFR
Timer 2 Clear Timer on Compare Match	TIFR
External Interrupt 1	GIFR

Question 7

5 pts

Given the USART initialization subroutine for the ATmega32 below. Determine the value of UBRRL for a baud rate of 19200 bps from the choices below. Assume fosc = 16 MHz. Round your answer to the nearest integer.

```
void usart_init(void) {
        UCCRA = (1<<U2X);
        UCSRB = (1<<TXEN) | (1<<TXCIE);
        UCSRC = (1<<UCSZ1) | (1<<UCSZ0) | (1<<URSEL);
        UBRRL = _____;
}
```

- ○ None of the above
- ○ 51
- ◉ 103
- ○ 52

Question 8

2 pts

Given the following C program. Fosc = 8MHz.

```
#INCLUDE <AVR\IO.H>
#INCLUDE <AVR\INTERRUPT.H>
ISR(TIMER0_COMP_vect);
void main()
{
        DDRA |= 0B00001000;
        TIMSK = (1<<OCIE0) ;
        TCNT0 = 0;
         OCR0 = 39;
        TCCR0 = 0X0B;
        SEI();
        WHILE(1);
}
ISR(TIMER0_COMP_vect)
{
        PORTA ^= 0X08;
}
```

True/False. The interrupt source for Timer 0 overflow is used in the code above.

- ○ True
- ◉ False

Question 9

2 pts

Given the following C program. Fosc = 8MHz.

```
#INCLUDE <AVR\IO.H>
#INCLUDE <AVR\INTERRUPT.H>
ISR(TIMER0_COMP_vect);
void main()
{
        DDRA |= 0B00001000;
        TIMSK = (1<<OCIE0) ;
        TCNT0 = 0;
         OCR0 = 39;
        TCCR0 = 0X0B;
        SEI();
        WHILE(1);
}
ISR(TIMER0_COMP_vect)
{
        PORTA ^= 0X08;
}
```

True/False. The waveform generated by this program toggles Port A bit 3.

- (●) True
- ( ) False

Question 10

5 pts

Given the following C program. Fosc = 16MHz. Answer the following questions.

```
#INCLUDE <AVR\IO.H>
#INCLUDE <AVR\INTERRUPT.H>
ISR(TIMER0_COMP_vect);
void main()
{
        DDRA |= 0B00001000;
        TIMSK = (1<<OCIE0) ;
        TCNT0 = 0;
        OCR0 = 39;
        TCCR0 = 0X0B;
        SEI();
        WHILE(1);
}
ISR(TIMER0_COMP_vect)
{
        PORTA ^= 0X08;
}
```

What is the period (not half-period) of the square wave generated in this program? Exclude the overhead in the period time calculation.

- ○ 640 us
- ○ 5 us
- ◉ 320 us
- ○ 10 us
- ○ 160 us

Question 11

2 pts

Given the C program program for serial communication. Answer the question below.

```
#include <avr/io.h>
unsigned char cleared = 1<<UDRE;

int main()
{
    UCSRB = (1<<RXEN) | (1<<TXEN);
    UCSRC = (1<<UCSZ1) | (1<<UCSZ0) | (1<<URSEL);
    UBRRL = 0x31;
    while(1)
    {
        while(!(UCSRA & cleared));
        UDR = 'D';
    }
    return 0;
}
```

True/False. Serial transmission is performed in the program above.

- (●) True
- ( ) False

Question 12

5 pts

Given the C program program for serial communication. Answer the question below.

```
#include <avr/io.h>
unsigned char cleared = 1<<UDRE;

int main()
{
    UCSRB = (1<<RXEN) | (1<<TXEN);
    UCSRC = (1<<UCSZ1) | (1<<UCSZ0) | (1<<URSEL);
    UBRRL = 0x31;
    while(1)
    {
        while(!(UCSRA & cleared));
        UDR = 'D';
    }
    return 0;
}
```

Determine the baud rate for the serial operation performed above. Use fosc = 16MHz.

- ○ 10000 bps
- ○ None of the above
- ◉ 20000 bps
- ○ 40000 bps
- ○ 2000 bps